

Exploring SARIMA and LSTM to forecast 24-hour Day ahead RT_Demand

Project by Isabella Lapsely and Ramson Munoz Morales

Outline

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 - Stationarity
 - Seasonality
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- **Results and comparison**
- **Conclusions**
- **Limitations**
- **Questions**

Research Problem

1. Can we forecast RT_demand 24 hours before it happens?

Why does this matter?



Research Problem

Why does this matter?

At 10 am today National grid submits a bid to ISO to supply 3000 MW of energy at 4pm tomorrow, 3500 MW at 5pm, etc. for all 24 hours of the next day.



Research Problem

Why does this matter?

If they order too much, they sell back to ISO at real time. Too little, and they have to buy more. A good estimate of energy demand means real money for the business at scale!



Background on data

- ISO - NE: Manages power distribution and the wholesale energy market for the New England area
- ISO - NE: Plans new energy grid expansion
 - The organization claims to be non-profit and independent
 - Provide data publicly

Data: Covered SEMA and CA NEMA from the New England Region, with data segment by each region as well as collective , for collected weather data weighted values were made of many weather stations in the region [1]

Dimensions Post Cleaning: 61,320 observations x 13 predictors



Fig. and data sourced from [1]

Learning from mistakes

- Added more data
- Fixing over fitting
- Exploring forecasting & time series methods
- Better understanding of the variables
- Research question better aligned with business goals

Monthly Energy Usage in New England

An analysis on ISO-NE data

Primary Question: What factors best predict monthly energy usage in ISO-NE in 2024?

By Hansel Frias, Edward Hutchens, Ramson Munoz Morales

Limitations & Future Directions

Limitations:

- Data is limited to monthly resolution
- Only 1 year of data
- For a given region we only have 12 data points

Future Directions:

- Expanding years analyzed
 - Monthly data since 2016
- Changing Resolution
 - Hourly vs Monthly vs Yearly
- Peak Demand vs Energy
 - Different stakeholders may care about different metrics

Data Pre-processing

We started with hourly data from 2016 to 2025

But the way that RT Demand is measured changed mid-way through 2018. Since we have a lot of hourly observations, we decided to consider 2019 through 2025—dropping the first 3 years worth of hourly observations.

Daylight savings reporting standard shifted in 2024

These were duplicate values that were dropped.

Variable Description

Variable	Category	Type	Description	Min	Q1	Med	Mean	Q3	Max	Use in model
RESPONSE VARIABLE										
RT_Demand (MW)	Demand — Real-Time (actual)	Continuous, hourly	ISO NE CA system load from revenue-quality metering: metered generation + net interchange + pumped-storage demand. Includes grossed-up demand response from Jun 2018.	1666	2297	2606	2661	2939	5491	RESPONSE
PREDICTORS										
DA_Demand (MW)	Demand — Day-Ahead (market-cleared)	Continuous, hourly	Cleared day-ahead demand: fixed bids + price-sensitive bids + net cleared virtuals. Market-based forecast of RT_Demand, not an analyst estimate.	1,366.00	2,273.00	2,586.00	2,635.00	2,916.00	5,197.00	DA (leakage — forecast of response)

Variable Description

Variable	Category	Type	Description	Min	Q1	Med	Mean	Q3	Max	Is used in model?
PREDICTORS										
DA_LMP (\$/MWh)	Price — Day-Ahead (LMP)	Continuous, hourly	Day-Ahead Locational Marginal Price, load zone (ISO NE CA tab = Trading Hub).	6.34	24.34	34.85	47.93	55.46	481.78	No because Price + DA
DA_EC (\$/MWh)	Price — Day-Ahead (energy component)	Continuous, hourly	Energy component of DA_LMP, load zone.	6.33	24.07	34.36	47.31	54.72	473.74	No because Price + DA
DA_CC (\$/MWh)	Price — Day-Ahead (congestion component)	Continuous, hourly	Congestion component of DA_LMP, load zone.	-10.18	0.00	0.00	0.10	0.01	114.45	No because Price + DA
DA_MLC (\$/MWh)	Price — Day-Ahead (marginal loss)	Continuous, hourly	Marginal Loss component of DA_LMP, load zone.	-1.80	0.22	0.37	0.52	0.62	7.89	No because Price + DA
RT_LMP (\$/MWh)	Price — Real-Time (LMP)	Continuous, hourly	Real-Time LMP, load zone; hourly average of the five-minute LMPs.	-151.05	22.19	33.22	46.81	54.09	2,274.99	No because Price
RT_EC (\$/MWh)	Price — Real-Time (energy component)	Continuous, hourly	Energy component of RT_LMP, load zone.	-151.17	21.93	32.81	46.23	53.40	2,239.32	No because Price

Variable Description

Variable	Category	Type	Description	Min	Q1	Med	Mean	Q3	Max	Use in model
PREDICTORS										
RT_CC (\$/MWh)	Price — Real-Time (congestion component)	Continuous, hourly	Congestion component of RT_LMP, load zone.	-8.78	0.00	0.00	0.05	0.01	67.69	No because Price
RT_MLC (\$/MWh)	Price — Real-Time (marginal loss)	Continuous, hourly	Marginal Loss component of RT_LMP, load zone.	-1.74	0.19	0.34	0.53	0.62	43.30	No because Price
Dry_Bulb (°F)	Weather	Continuous, hourly	Zone-weighted dry-bulb temperature. Summer Jun–Sep, Winter Oct–May (see station weights).	-9.00	40.00	53.00	53.42	67.00	101.00	
Dew_Point (°F)	Weather	Continuous, hourly	Zone-weighted dewpoint temperature. Same summer/winter convention.	-31.00	27.00	42.00	40.81	56.00	77.00	
Date_Time	Time index	POSIXct (hourly)	Combined timestamp; used as the series index.	2019-01-01 00:00	2020-10-01 05:15	2022-07-02 10:30	2022-07-02 10:52	2024-04-01 16:45	2025-12-31 23:00	Index

Time series: What makes it more complicated?

Typical Regression techniques

- Observations are independent
- Predictors are other variables
- Does not require stationarity

$$\mathbf{y} = \beta_0 \mathbf{x}_0 + \sum_{i=0}^n \beta_i \mathbf{x}_i + \epsilon$$

Time series - SARIMA example

- Observations are auto-correlated
- The response variable at an earlier time is a predictor
- Requires stationarity

$$By_t = y_{t-1}, \quad B^k y_t = y_{t-k}$$

Time series: Autocorrelation

$$I_t \approx I_{t-1}$$

Yesterday says a lot of the same things
today says :

$$SE = \frac{\sigma}{\sqrt{n}}$$

Overly confident:

If you have 7 instances of someone's day and key indicators to track their happiness :

5-week days, all very similar (autocorrelated)

2- weekend days also similar (autocorrelated),

7 instances are not all providing novel information on how these metrics influence a person's happiness .

The actual data is more like only two instances

Some things like to stay the same :

Misleading:

Let's say people's state of mind is very stable (#zen) : If they are happy today very little will change in their happiness tomorrow (auto correlated)

This means the influence of the indicators tracked ,that the model is trying to learn in relation to the final happiness metric, is including how zen people are in addition to the actual influence of the variables .

Bad weeks are BAD but not so much because of the indicators but because it's been a rough few days (the previous instances are similar intrinsically)

A bad day on a good week not so bad -> model needs the previous days to account for that enter in the Lag variable

$$\begin{aligned} \text{😊}_t &= \beta_0 + \beta_1 X_t + \dots + \epsilon_t \\ \text{😊}_t &= \beta_0 + \beta_1 X_t + \dots + \beta_i \text{😊}_{t-1} + \epsilon_t \end{aligned}$$

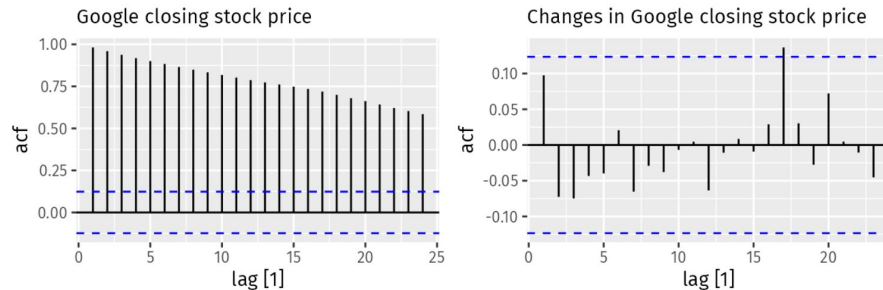
Time series: Stationarity

A stationary time series is one whose statistical properties do not depend on the time at which the series is observed (Hyndman et. al, 2021)

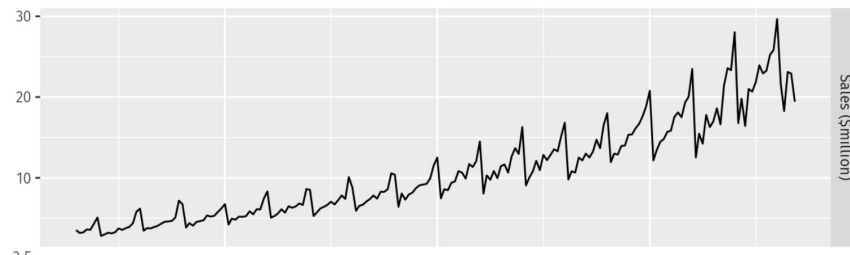
- Trends in the data break stationarity
- Seasonality breaks stationarity
- Cycles do not break stationarity

Images sourced from
<https://otexts.com/fpp3/stationarity.html>
 (Hyndman et. al., 2021)

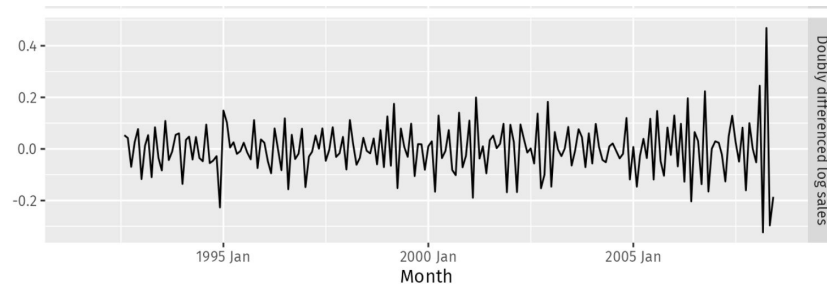
Not Stationary



Antidiabetic drug sales



Stationary



Time series: Seasonality

- Patterns of rises and falls in a data that follow a fixed pattern.
 - In the heatmap to the top right, RT_Demand increases from July to September every year and December to March.
 - Additionally, RT_Demand falls in the autumn and spring months.
- Cycles are repeated rise and fall without a fixed time pattern
 - In the line chart to the bottom right, there is a rise and fall, but the width of the peaks and valleys changes making cycles not seasons

Yearly RT_Demand by Week

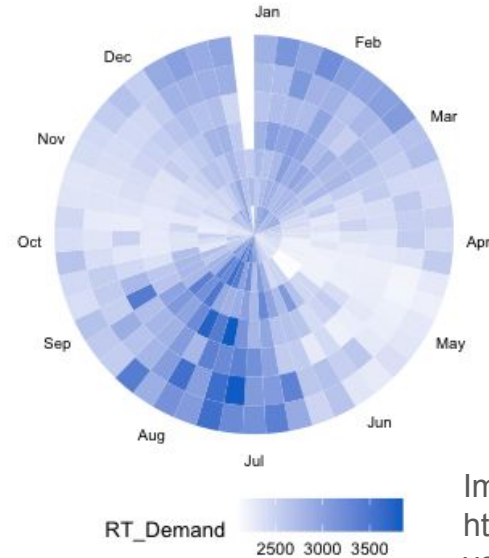
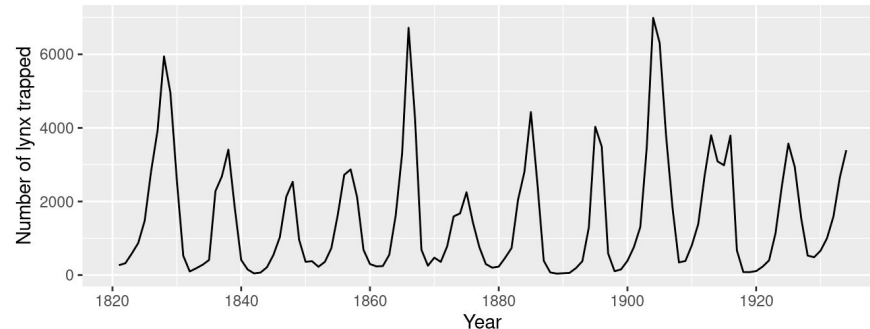


Image below sourced from <https://robjhyndman.com/hyndsight/cyclights/>

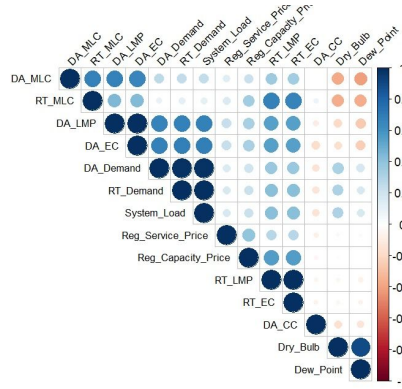
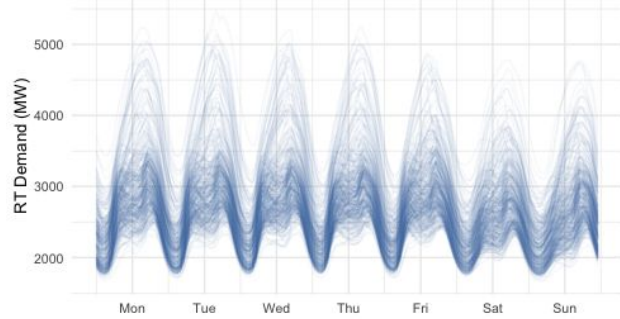


Exploratory Data Analysis

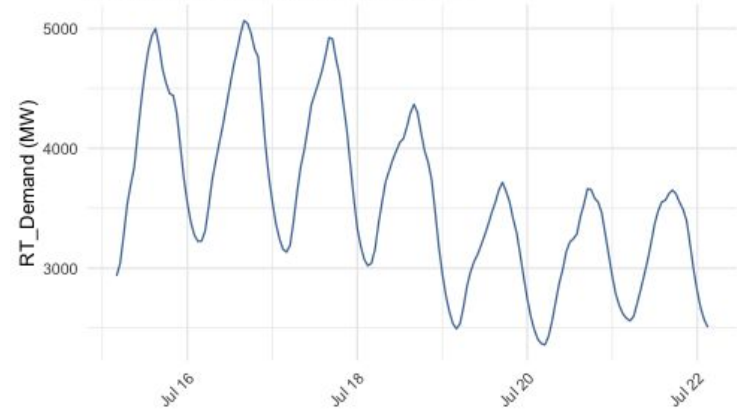
- Literal transition slide

All weeks overlaid, 2019–2025

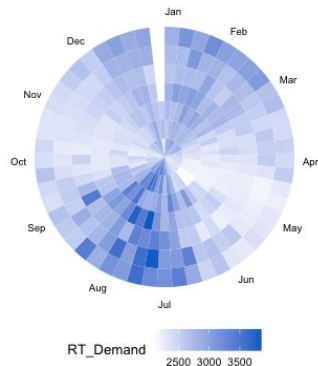
Each line is one week (168 hourly points)



Hourly demand, week of Jul 15 2024

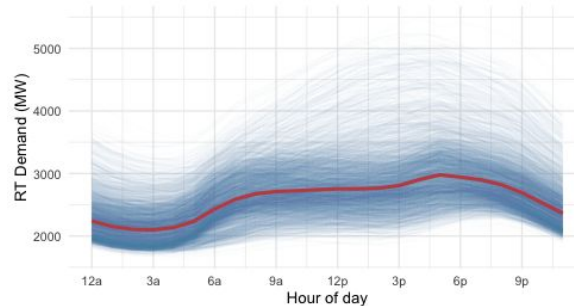


Yearly RT_Demand by Week

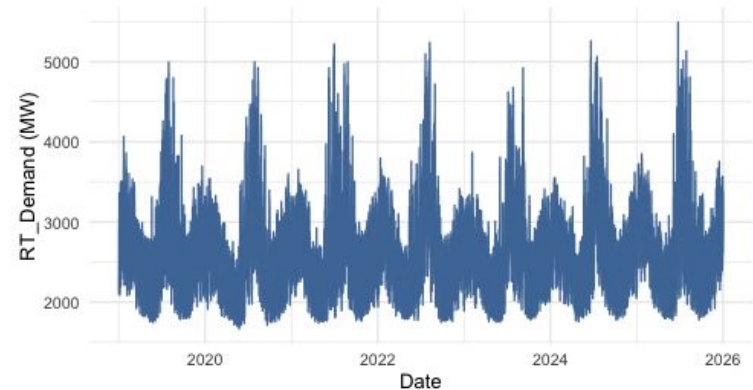


Daily demand cycle — all days overlaid, 2019–2025

Blue: individual days · Red: median across all days



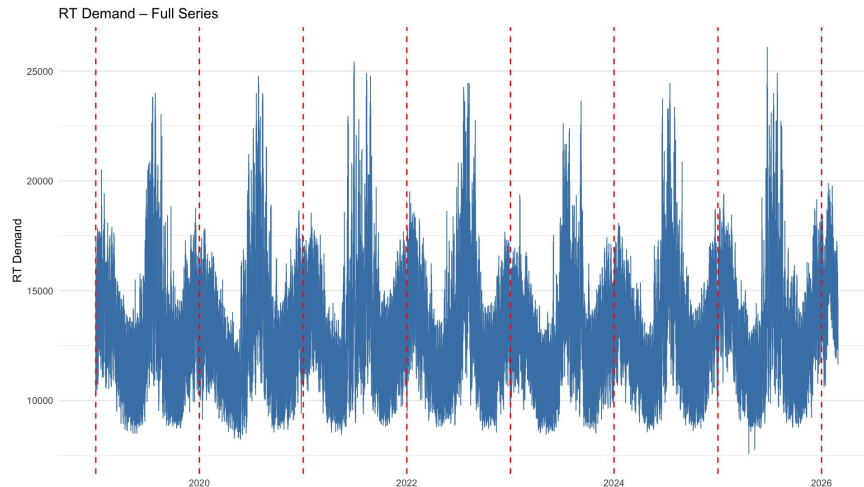
Real-time demand by hour of day
ISO New England, 2019-2025



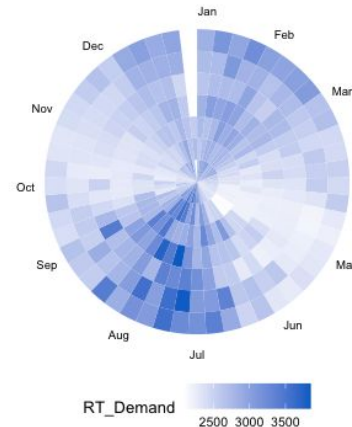
Exploratory Data Analysis

At the yearly scale

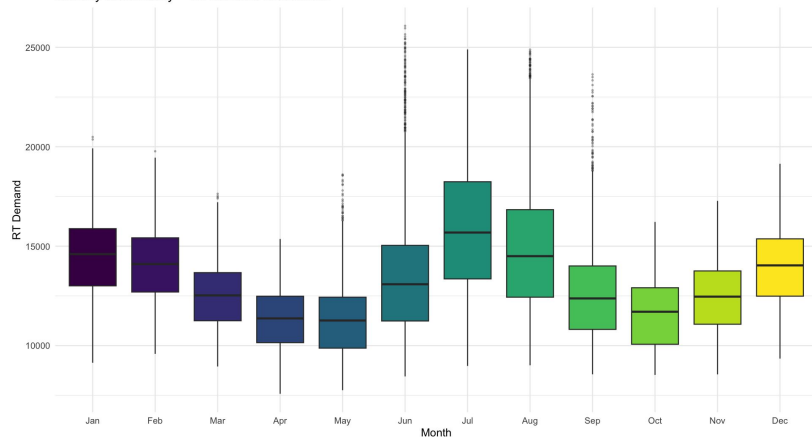
- Seasonality indicated in heatmap
- Seemingly stable to trends



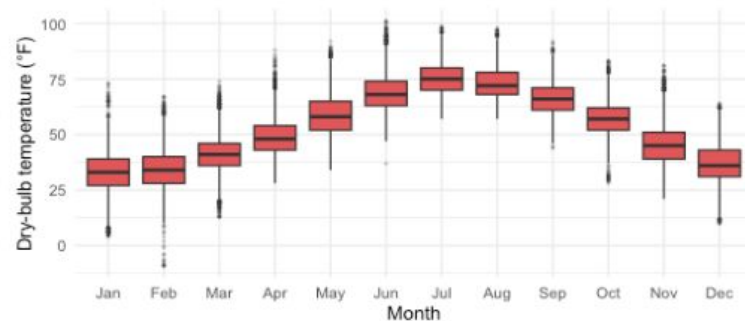
Yearly RT_Demand by Week



Monthly Seasonality – RT Demand Distribution



Temperature by month

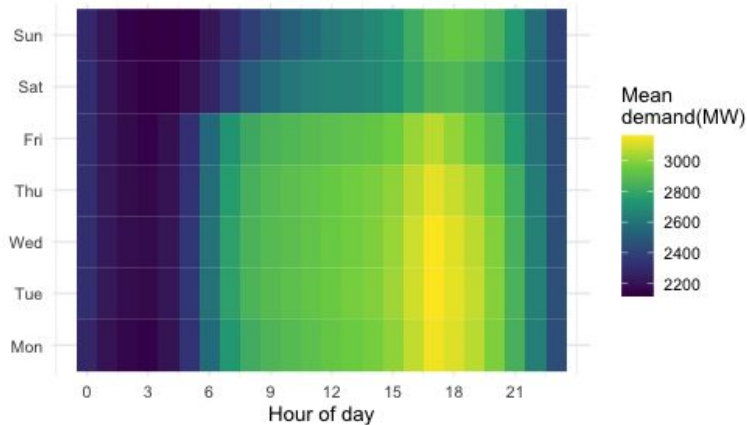


Exploratory Data Analysis

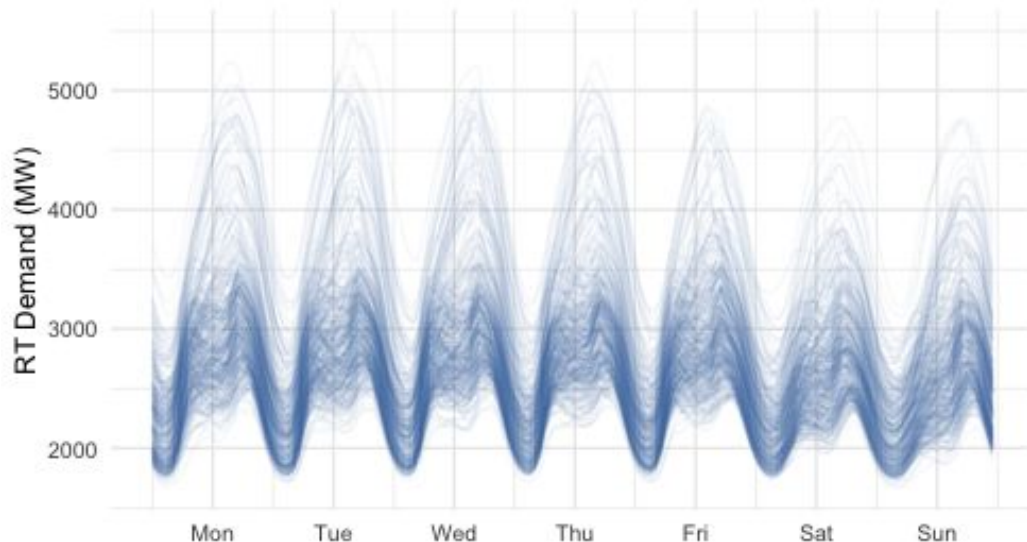
At the weekly scale

- Weekly seasonality present
- 24 Hour seasonality indicated
- Weekday vs weekend trend

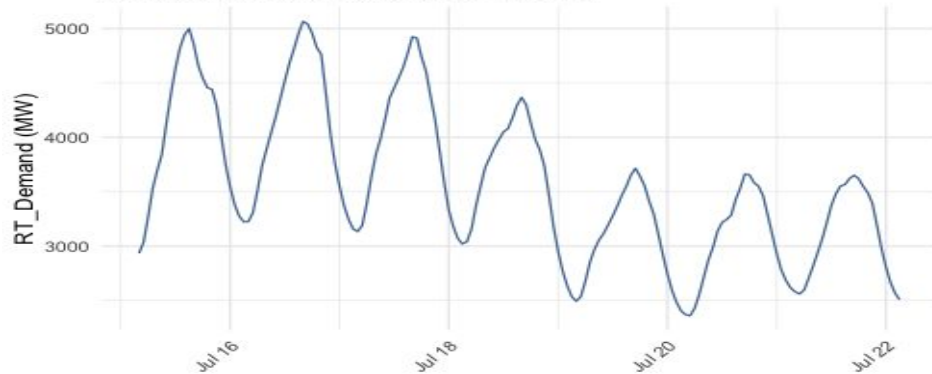
Mean demand by day of week and hour of day



All weeks overlaid, 2019–2025
Each line is one week (168 hourly points)

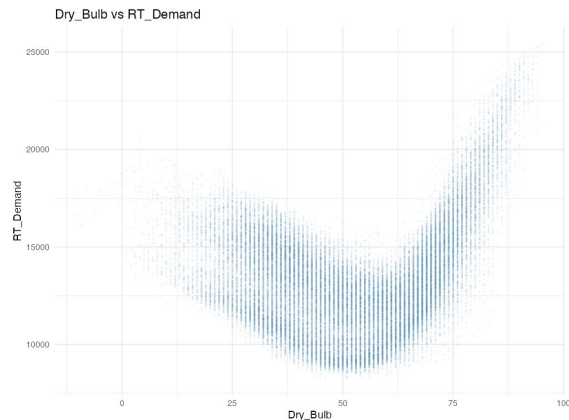
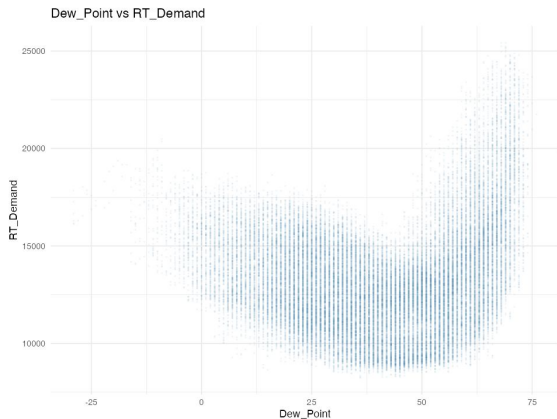


Hourly demand, week of Jul 15 2024



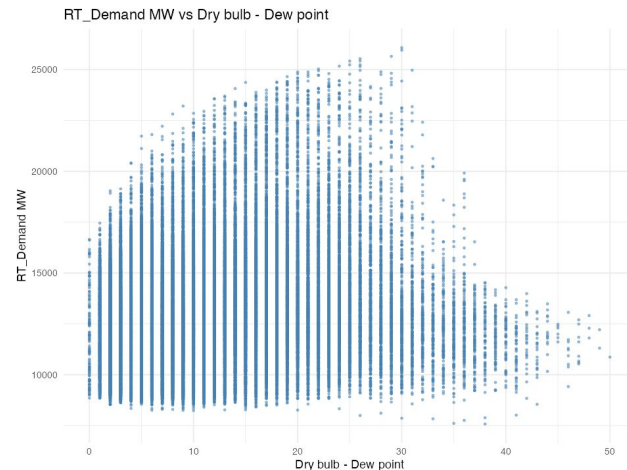
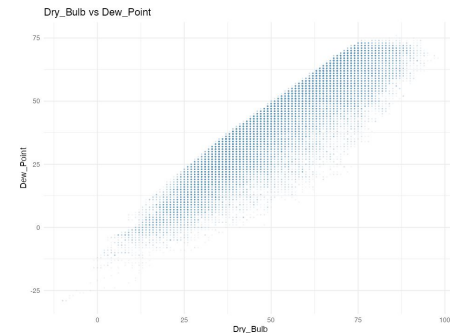
Exploratory Data Analysis

Dew point and Dry Bulb Temp



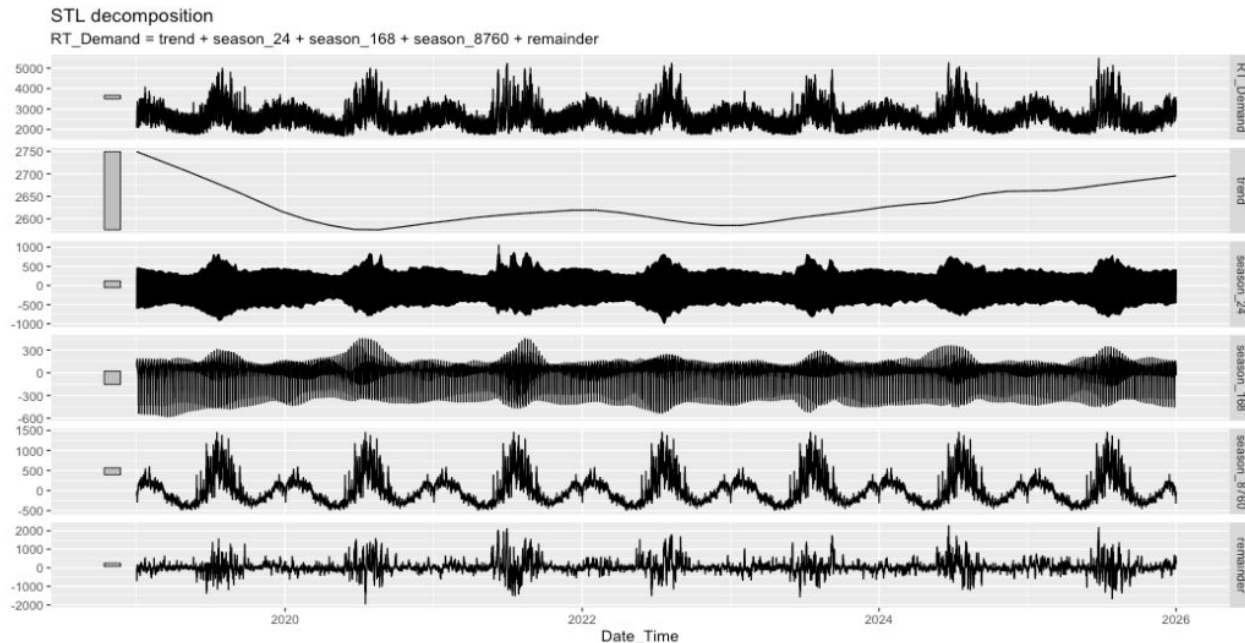
Dew Point: is the temperature at which water vapor starts to condense out of the air. It gives humidity when in relation to Dry bulb, if they are close it is humid .[7]

Dry bulb : the ambient air temperature “When people refer to the temperature of the air, they are normally referring to its dry bulb temperature.”[7]



Why SARIMA instead of ARIMA?

- The series is stationary with respect to trends
- But we have three clear seasonal periods at yearly, weekly, and daily scale
- A multi-season model may be better (prophet, TBATS)



Trend stationary at 95% confidence

Augmented Dickey-Fuller Test

```
data: project_data$RT_Demand
Dickey-Fuller = -16.664, Lag order = 39, p-value = 0.01
alternative hypothesis: stationary
```

KPSS Test for Level Stationarity

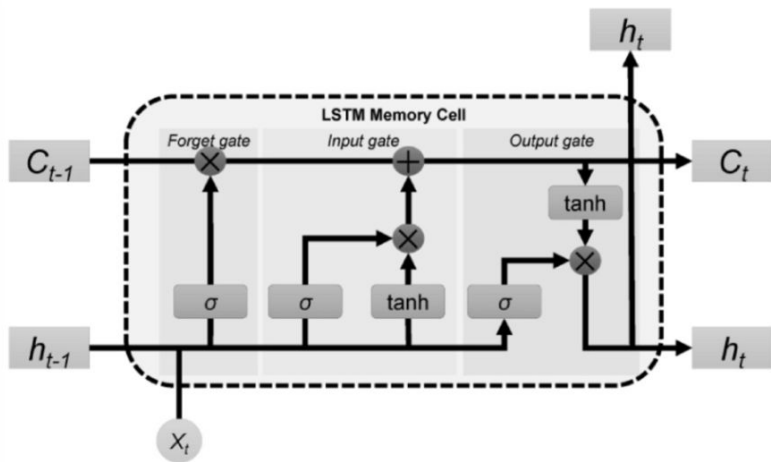
```
data: project_data$RT_Demand
KPSS Level = 0.61614, Truncation lag parameter = 19, p-value = 0.02117
```

Seasonal **A**uto**R**egressive **I**ntegrated **M**oving **A**verage

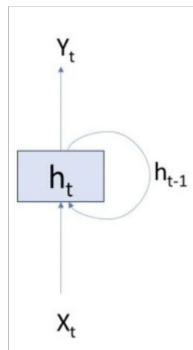
$$\underbrace{\Phi_P(B^m)}_{Pterms} \underbrace{\phi_p(B)}_{pterms} \underbrace{(1-B)^d}_{dtimes} \underbrace{(1-B^m)^D}_{Dtimes} y_t = \underbrace{\Theta_Q(B^m)}_{Qterms} \underbrace{\theta_q(B)}_{qterms} \varepsilon_t$$

- p - autoregression is how many previous values to include in the model
 - Example: RT_Demand from 4pm to predict RT_Demand at 5pm (p = 1). Or, using the RT_Demand from 3 pm and 4 pm to predict RT_Demand at 5pm (p=2)
- P - autoregression on a seasonal scale
 - Example: RT_Demand from 5pm yesterday to predict RT_Demand at 5pm today (P=1)
- d - differencing (the “I” in SARIMA) we assume $\nabla y_t = y_t - y_{t-1}$ and $(1-B)^d$ captures the amount of differencing needed to remove trends in the data.
 - Proof not shown here. But effectively the power of the difference grows
- D - differencing on seasonal scale
- q - moving average of the error from the last value considered
 - Example: error from 4pm is included in the prediction for 4pm (q=1)
- Q - moving average of the error from the last seasonal period
- m - the seasonal period **NOTE: SARIMA only allows for one seasonal period**

Traditional Long Short Terms Memory and why they might want to be changed



A type of recurrent neural network designed to overcome the vanishing gradient problem. It achieves this by splitting memory into two components. A cell state that is a long-term memory and the previous state (a stand-in for short-term memory). This architecture provides a more direct gradient flow path through time, allowing the network to retain relevant historical information without gradients decaying or exploding over time across many time steps.



Traditional LSTM are deterministic

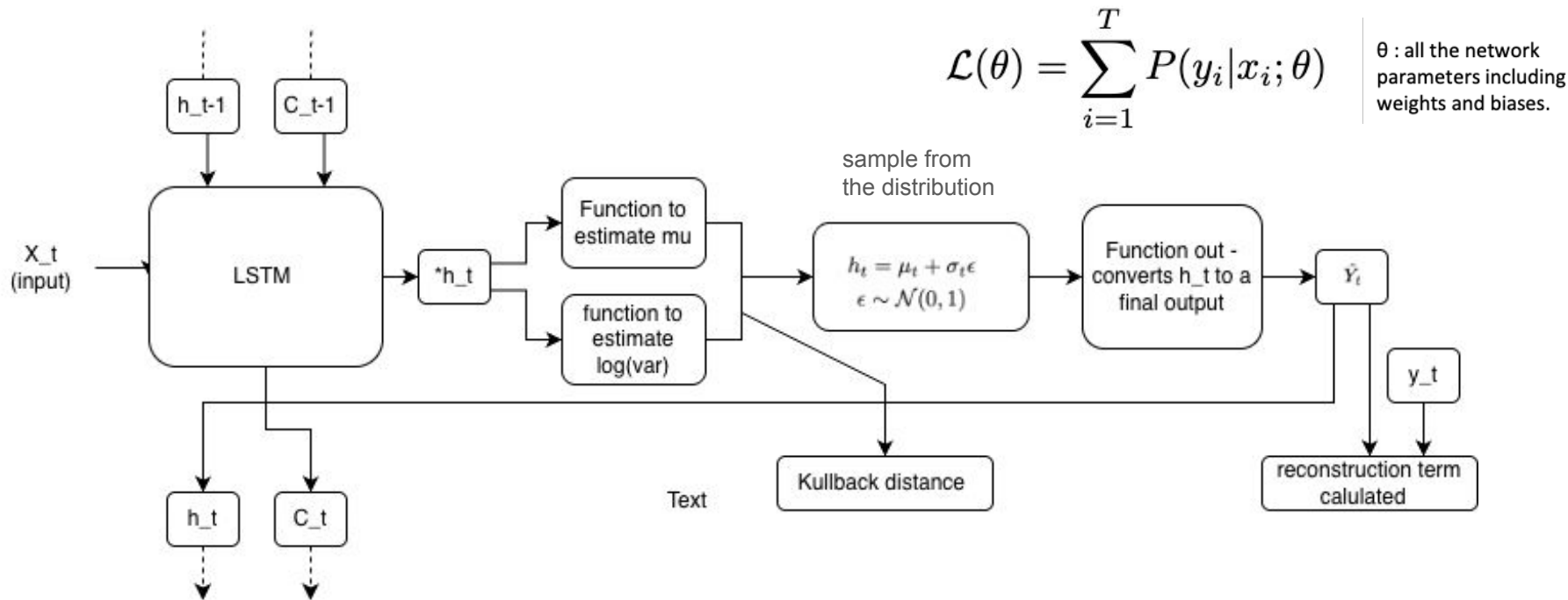
This means that there is no probability distribution, so traditional statistical methods for model selections like AIC that helps with input and parameter selections would not be possible

A stochastic LSTM could change this

In a stochastic model the hidden states are probabilistic, changing the model to work on probabilities making input and parameter selections with traditional statistical methods possible.

Additionally, stochastic variables will introduce “stochastic noise” into the training process. Instead of the inputs that always produce the same hidden state values, you get more variety that reduces the risks over overfitting. The stochastic nature of the model works as a regularizing force.

Implementation of a Stochastic LSTM -simplified

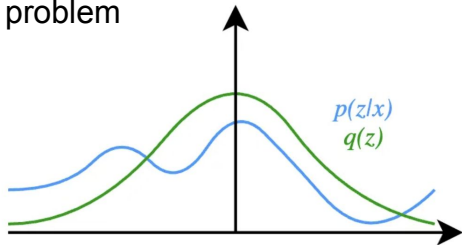


The loss function used to update the gradients

$$\mathcal{L}(h_t, \theta, \phi) = \text{reconstruction}_{t-1} + (-.5 \log(2\pi) - .5(y_t - \hat{y})^2) - (KL_{t-1} + 0.5(\mu_q^2 + \sigma_q^2 - 1 - \log(\sigma_q^2))) + \lambda\Omega(\theta) + \lambda_{\text{hidden}}\Omega_{\text{hidden}}(\theta)$$

How do you derive the likelihood

Variational inference : is a group of methods that all approximates a true posterior distribution (in this case $p(h_{1:T} | y_{1:T}, x_{1:T}; \theta)$) with a tractable variational distribution $q(h_{1:T})$. Generally taking advantage of a result of Kullback-leibler distance, it makes a complicated integration problem an optimization problem



We will assume that $q(h_t)$ represented as $q(z)$ in the figure is a gaussian distribution

fig (Chen, 2025)[4]

$P(Y_{1:t}|X_{1:t} : \theta)$ is actually $P(Y_{1:t}|h_1, h_2, \dots, h_t, X_{1:t} : \theta)$
marginalizing you get $\int_{h_1} \int_{h_1} \dots \int_{h_t} P(Y_{1:t}|X_{1:t} : \theta)$ This is intractable):

$$P(Y_{1:t}|X_{1:t}; \theta) = \frac{P(Y_{1:t}, h_{1:t}|X_{1:t}; \theta)}{P(h_{1:t}|Y_{1:t}X_{1:t}; \theta)} \quad \text{from 1 joint distribution formula}$$

$$\log(P(Y_{1:t}|X_{1:t})) = \log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(P(h_{1:t}|Y_{1:t}X_{1:t}))$$

Say we estimate the posterior probability $P(h_{1:t}|Y_{1:t}X_{1:t})$ with some distribution $q(h_{1:t})$
 $\log(P(Y_{1:t}|X_{1:t})) = \log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(P(h_{1:t}|Y_{1:t}X_{1:t})) + \log(q(h_{1:t})) - \log(q(h_{1:t}))$

$$\log(P(Y_{1:t}|X_{1:t})) = \log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(q(h_{1:t})) + \log\left(\frac{q(h_{1:t})}{P(h_{1:t}|Y_{1:t}X_{1:t})}\right)$$

$$E_{q(h_{1:t})} [\log(P(Y_{1:t}|X_{1:t}))] = E_{q(h_{1:t})} [\log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(q(h_{1:t}))] + E_{q(h_{1:t})} [\log\left(\frac{q(h_{1:t})}{P(h_{1:t}|Y_{1:t}X_{1:t})}\right)]$$

$$E_{q(h_{1:t})} [\log\left(\frac{q(h_{1:t})}{P(h_{1:t}|Y_{1:t}X_{1:t})}\right)]: \text{ Is the Kullback-Leibler distance } KL(q(h_{1:t})||P(h_{1:t}|Y_{1:t}X_{1:t}))$$

Kullback-Leibler distance is a measure of the information loss when estimating a distribution with another

$$\log(P(Y_{1:t}|X_{1:t})) = E_{q(h_{1:t})} [\log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(q(h_{1:t}))] + KL(q(h_{1:t})||P(h_{1:t}|Y_{1:t}X_{1:t}))$$

$$\log(P(Y_{1:t}|X_{1:t})) - KL(q(h_{1:t})||P(h_{1:t}|Y_{1:t}X_{1:t})) = E_{q(h_{1:t})} [\log(P(Y_{1:t}, h_{1:t}|X_{1:t})) - \log(q(h_{1:t}))]$$

KL can only be 0 or higher so

$$\log(P(Y_{1:t}|X_{1:t}; \theta)) \geq E_{q(h_{1:t})} [\log(P(Y_{1:t}, h_{1:t}|X_{1:t}; \theta)) - \log(q(h_{1:t}; \theta))]$$

This becomes the objective function of the optimization problem: the evidence lower bound (ELBO)

:)

But how do you calculate the ELBO and also allow for gradient calculations required to update the model weights?

Key assumption:

$$P(h_t|h_{1:t-1}X_t; \theta) \sim \mathcal{N}(0, 1) \quad q(h_t; \phi) \sim \mathcal{N}(\mu_q, \sigma_q^2) \quad \phi = (\mu_q, \sigma_q^2) \quad p(y_t|h_t : \theta) \sim \mathcal{N}(W_y h_t + b_y, \sigma^2)$$

We can then rearrange the Elbo term in to two key components

$$\mathcal{L}(h_t, \theta, \phi) = E_{q(h_{1:t})}[\log(P(Y_{1:t}|h_{1:t}, X_{1:t}; \theta))] - KL(q(h_{1:t}; \phi) || (P(h_{1:t}|X_{1:t}; \theta))$$

Reconstruction term : $E_{q(h_{1:t})}[\log(P(Y_{1:t}|h_{1:t}, X_{1:t}; \theta))]$ Kullback distance

$$\mathcal{L}(h_t, \theta, \phi) = \prod_{t=1}^T (\log P(Y_t|h_t, X_t; \theta) P(h_t|h_{1:t-1})) - \sum_{1-t}^T KL(q(h_t; \phi) || (P(h_t|X_{1:t}; \theta)))$$

So at any given t:

$$\mathcal{L}(h_t, \theta, \phi) = \text{reconstruction}_{t-1} + (-.5 \log(2\pi) - .5(y_t - \hat{y})^2) - (KL_{t-1} + 0.5(\mu_q^2 + \sigma_q^2 - 1 - \log(\sigma_q^2)))$$

This is a butchering of the Variational Autoencoder (VAE) with Long Short-Term Memory (LSTM) which would use a separate LSTM model to “encode” $P(h_t|h_{1:t-1}X_t; \theta)$ [5]

Lasso regression applied to LSTM

L1 Regularization (Lasso Penalty): Penalizes the ℓ_1 -norm of the weights

$$\mathcal{L}(\theta) = -\log p(y_{1:T} \mid x_{1:T}; \theta) + \lambda \Omega(\theta) + \lambda_{\text{hidden}} \Omega_{\text{hidden}}(\theta),$$

$$\Omega(\theta) = \sum_{g \in \{i, f, o, c\}} (\alpha_1 \|W_g\|_1 + \alpha_2 \|U_g\|_1) + \alpha_3 \|b_g\|_1, \quad \Omega_{\text{hidden}}(\theta) = \beta \sum_{t=1}^T \|h_t\|_1,$$

Shrinkage on :

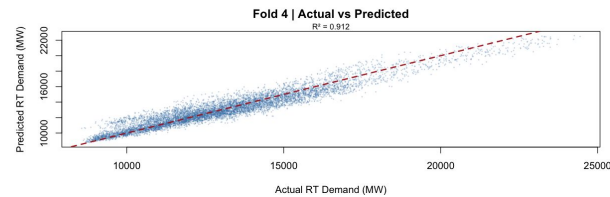
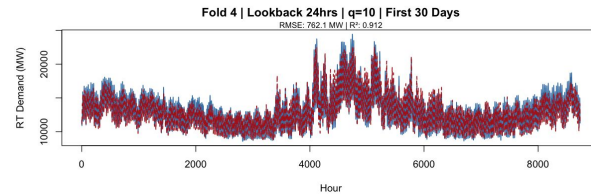
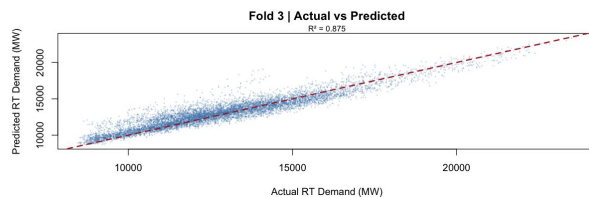
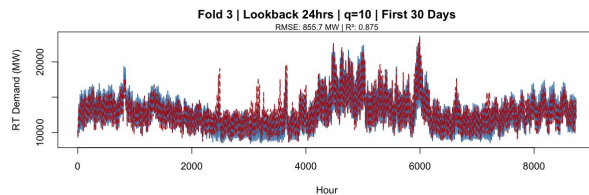
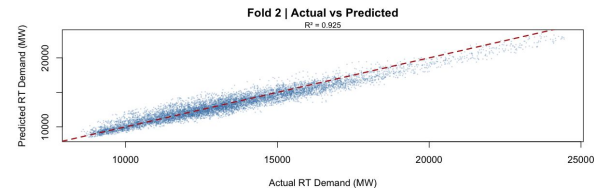
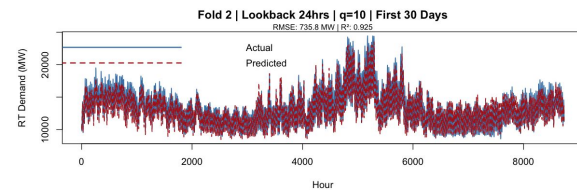
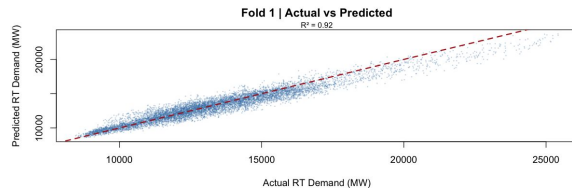
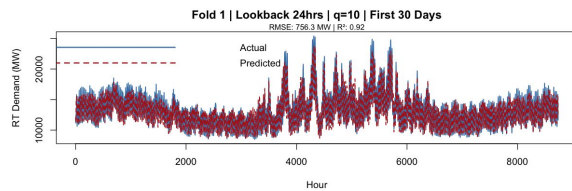
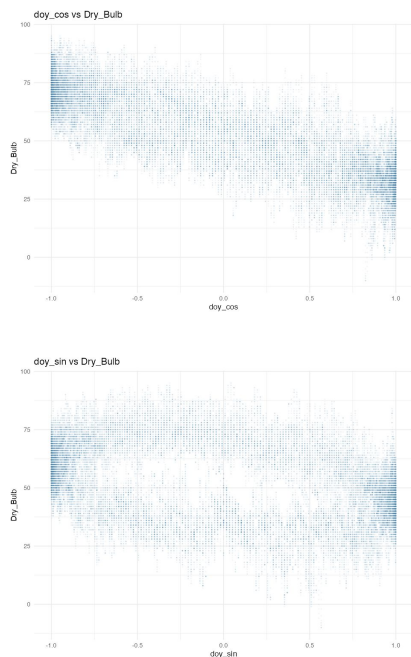
Wg : input feature selection, asses which external inputs are most influential for a specific gate.

Ug: encourages sparsity in the recurrent connections, controlling memory dynamics over time.

Bg: ensures that biases do not lead to unbalanced gate activations without strong evidence.

Equ. and text sourced from Mostafa, F. (2025) [1]

Results-LSTM



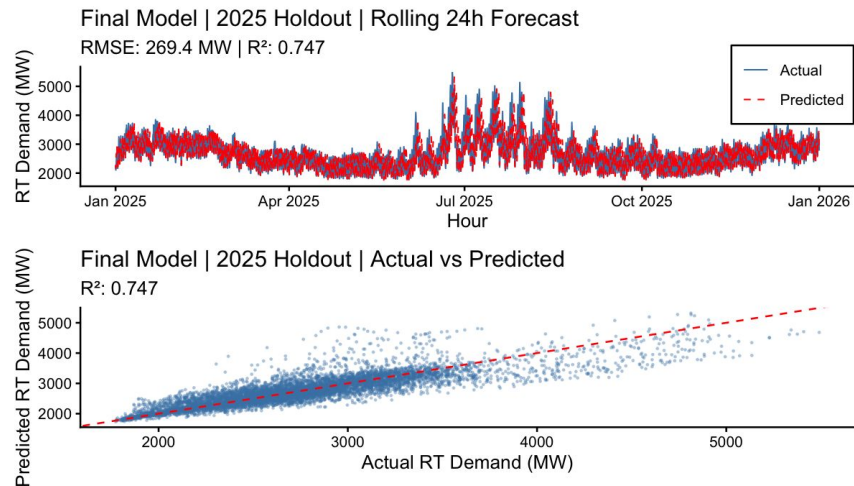
Fold	Fold #	Lookback	Best Q	Selected Features	AIC	Test RMSE
1_24	1	24	10	Dry_Bulb, hour_cos, dow_cos, month_cos, doy_sin	-191586.4	756.2947
2_24	2	24	10	Dry_Bulb, Dew_Point, hour_sin, hour_cos, dow_sin, dow_cos, month_cos	-192378.3	735.7817
3_24	3	24	10	Dry_Bulb, hour_cos, dow_cos, month_cos	-191445.4	855.7197
4_24	4	24	10	Dry_Bulb, hour_cos, dow_sin, dow_cos, doy_sin, doy_cos	-191789.0	762.0899

AIC-based Model selection with Cross Validation

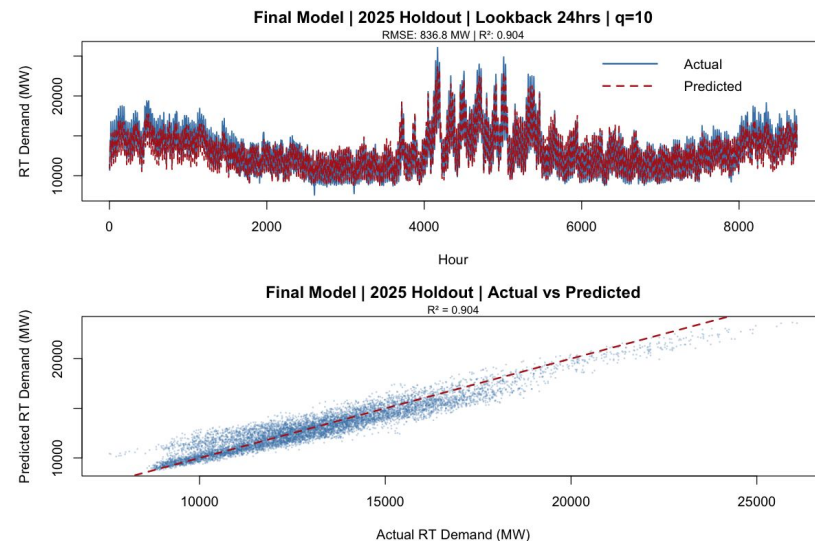
- Cross validation set up via sliding window and expanding window
- Cross validation estimated AIC as a model selection process
- A holdout of the hourly observations for 2025 was retained for head to head model comparisons
- Both models were refit on the full training data using AIC then tested on the holdout

Results on test set 2025

SARIMA



LSTM



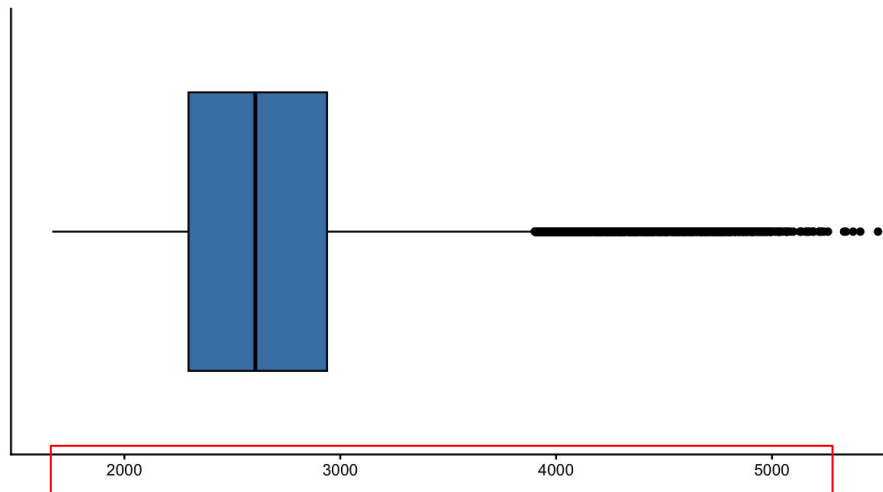
Final selected features: Dry_Bulb, Dew_Point, hour_sin, hour_cos, dow_sin, dow_cos, month_sin, month_cos, doy_sin, doy_cos

Model	AIC	RMSE	R^2
SARIMA	606468.3	269.3981	0.747
LSTM	-639864.8	836.77	0.9037

“The best laid plans of mice and men often go awry...” - Steinbeck

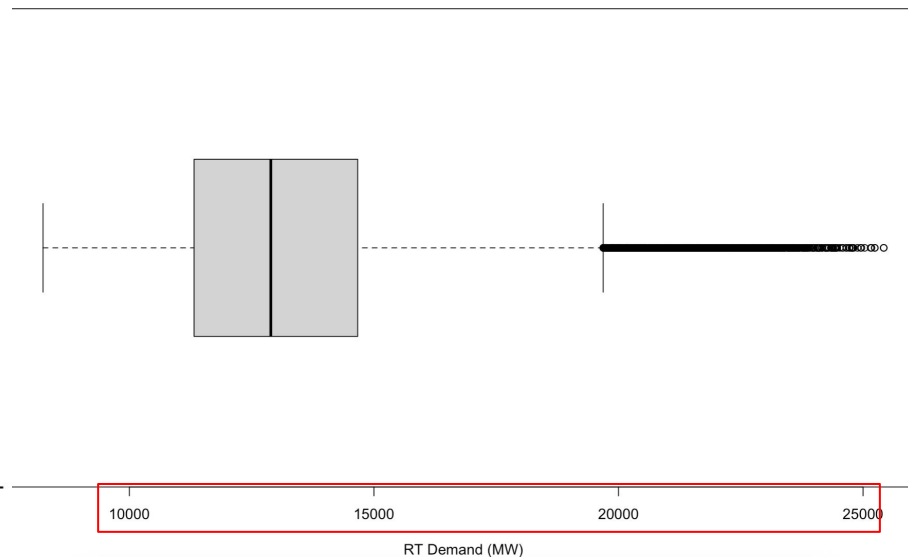
Note Differences in scale SEMA vs CA NEMA

Distribution of RT_Demand
SEMA 2019 - 2025



SEMA RT_Demand ranges from 1666 to 5491 MW

RT Demand Distribution (2019-2025)



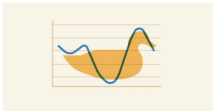
CA NEMA RT_Demand ranges from 7,577 to 26,081 MW

Conclusions

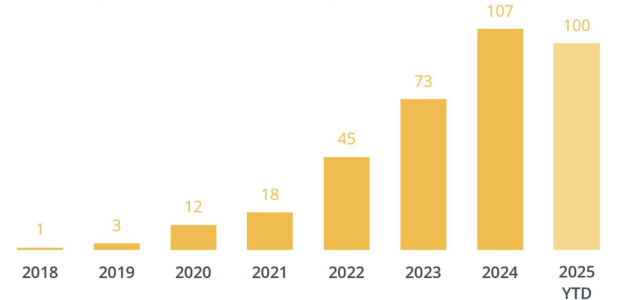
- Our plan for model comparison was to use the last year of data to compare model performance on the same task.
- However, when comparing R^2 and RMSE we noticed large discrepancies between the R^2 and RMSE.
- Investigation of the variance in the response variable revealed different ranges for the response.
- The consistency in the pattern we do see could potentially indicate that the seasonal behavior in RT_Demand is scale and region independent (within the New England area)

Limitations

- General limitations:
 - Range in look backs were never tested, limited to 24 hr input histories, may find use at different interval
 - Different data (we only found out yesterday)
 - Modeling fitting and assessment requires another 24-48 hours to produce estimates limiting our turnaround time.
- Limitations on SARIMA model:
 - Only accounted for 24 seasonal period despite weekly and yearly seasonal behavior present
 - Solar increasing popularity in 2018
 - Demand has nonlinear relationship with time.
 - Not accounted for in our model.
 - A known issue. [2]
- Limitations on lstm model:
 - Hardware and software issues leading to very aggressive assumptions that end up countering the very nature of LSTM models .
 - No testing on lambda for the penalties , no cross validations was done to test different penalty levels (may still have room for improvement)



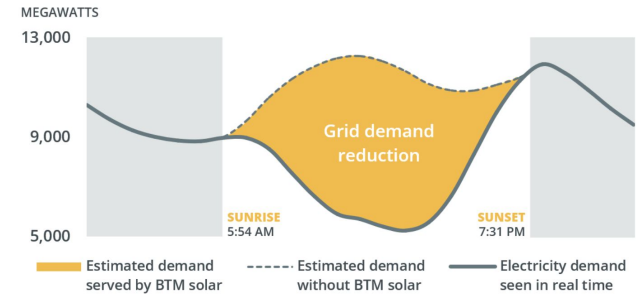
New England duck curve days per year



SOURCE: ISO NEW ENGLAND

[2]

Grid demand reduction from BTM solar, April 20



SOURCE: ISO NEW ENGLAND

Images sourced from ISO Newswire [2]

[2]

Questions?

References

- [1] ISO New England Inc. (2026). *Zonal Information* [Data set]. ISO Express Energy, Load, and Demand Reports. <https://www.iso-ne.com/isoexpress/web/reports/load-and-demand/-/tree/zone-info>
- [2] ISO New England Inc. (2025, September 17). Region's 100th 'duck curve' day migrates to early fall. *ISO Newswire*. <https://isonewswire.com/2025/09/17/regions-100th-duck-curve-day-migrates-to-early-fall/>
- [3] Mostafa, F. (2025). A Statistical Framework for Model Selection in LSTM Networks. arXiv preprint arXiv:2506.06840. <https://arxiv.org/abs/2506.06840>
- [4] Chen, Y. (2025, February 11). Introduction to Kullback-Leibler divergence. Medium. <https://medium.com/@yian.chen261/introduction-to-kullback-leibler-divergence-2d76979d1d8c>
- [5] Shende, R. (2023, April 19). Autoencoders, variational autoencoders (VAE) and β -VAE. *Medium*. <https://medium.com/@rushikesh.shende/autoencoders-variational-autoencoders-vae-and-beta-vae-ceba9998773d>
- [6] Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- [7] Haby, J. (n.d.). *Temperatures including dry bulb / wet bulb / dew point / potential / equivalent potential*. National Weather Service. Retrieved April 21, 2026, from https://www.weather.gov/source/zhu/ZHU_Training_Page/definitions/dry_wet_bulb_definition/dry_wet_bulb.html

Not directly referenced but vital to understanding

- Ganguly, A., Jain, S., & Watchareeruetai, U. (2023). Amortized variational inference: A systematic review. *Journal of Artificial Intelligence Research*, 78, 167–215. <https://doi.org/10.1613/jair.1.14258>
- Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518), 859–877. <https://doi.org/10.1080/01621459.2017.1285773>